Classifying EEG of Propofol-Induced Unconsciousness in the Presence of Burst Suppression

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***Abstract*—Anesthesia delivery yields dynamic electroencephalogram (EEG) signatures during moments of consciousness and unconsciousness. Consequently, real-time analysis of EEG signals during anesthesia administration is a promising technique to monitor patient states. Moreover, EEG has the potential to be the prime brain-monitoring technique for the creation of a closed-loop anesthesia delivery (CLAD) system. Machine learning (ML) software that can classify EEG time samples as conscious or unconscious is necessary for further CLAD system development. Burst suppression, a common metabolic phenomenon in deeply unconscious patients, presents a unique challenge to engineering an effective classifier. In this project, a previously engineered EEG classifier was replicated using band wise power (BWP) and principal component analysis (PCA) as features in a logistic regression classifier. The classifier performed well on the training data (BWP AUC = 0.94428, PCA AUC = 0.96096) and validation sets (BWP Mean AUC = 0.94426, SEM = 0.0010468, PCA Mean AUC = 0.96099, SEM = 0.0010535) but performed poorly when burst suppression samples were isolated for performance (BWP Hit rate = 0.4062, PCA Hit rate = 0.4330). Two new supervised ML approaches, support vector machines (SVM) and single-layer artificial neural networks (ANN), were utilized to improve classifier performance on burst suppression samples. As a result, the best performing classifier, PCA trained SVM, generated a burst suppression hit rate of 0.9598 on the training set and 0.8660 on the testing set while maintaining an AUC of 0.94714. While this project demonstrates that a classifier utilizing PCA with SVM is a promising avenue for robustly characterizing EEG signals in the presence of burst suppression, we conclude that dynamic preprocessing normalization techniques need to be implemented for optimal performance without overfitting.**

***Index Terms*—electroencephalogram (EEG) analysis, propofol, consciousness, closed-loop anesthesia delivery (CLAD), machine learning (ML), burst suppression, logistic regression (LR), band-wise power (BWP), principal component analysis (PCA), support vector machine (SVM), artificial neural network (ANN)**

I. Introduction

The anesthetic state is characterized by unconsciousness, amnesia, antinociception, and immobility with physiological stability [1]. Anesthesiologists are trained to monitor this state via multiple patient signs (blood pressure, heart rate, respiratory rate, ect.) and make necessary adjustments to drug delivery in response to changes in vitals. Notably, the use of real-time neuroimaging techniques for patient monitoring is not currently a part of the standard anesthesia practice. However, electroencephalogram (EEG) recordings are beginning to be used more widely in anesthesiology as physicians become more acclimated to reading time and time-frequency signals in the operating room [2].

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Recent industry developments have generated index readings from EEG frontal leads that can provide doctors a scalar value from 0 to 100 that serves as a marker of a patient's level of consciousness [3]. One such index, known as the bispectral index (BIS), has been utilized in the development of a closed-loop anesthesia delivery (CLAD) system (see [4], [5]), however it is prone to misleading measurements [6], [7]. This is in part due to the algorithm's insensitivity to the type of drugs in the anesthetic infusions. Comparisons between BIS and another commonly used index, the Wavelet-based Anesthetic Value for Central Nervous System (WAVCNS) index, show mismatches on the measurement of consciousness in patients [2]. These examples illustrate that index labeling of EEG signals as conscious or unconscious during anesthesia may not be an effective route for CLAD development.

Burst suppression is a special state of unconsciousness that generates idiosyncratic EEG signals in both the time domain and the time-frequency domain (Figure 1, [2]). The prevailing theory behind burst suppression involves complex interactions between dynamic cortical processes and overall brain metabolism [8]. These interactions result in an EEG signal characterized by alternating moments of transient bursting patterns and isoelectric activity in the time domain and universal power across frequencies in the time-frequency domain [2].

An effective classifier that can categorize EEG signals as conscious or unconscious is necessary for proper CLAD system development, and burst suppression offers a challenge to engineering such a system. Because burst suppression constitutes the deepest state of unconsciousness before a

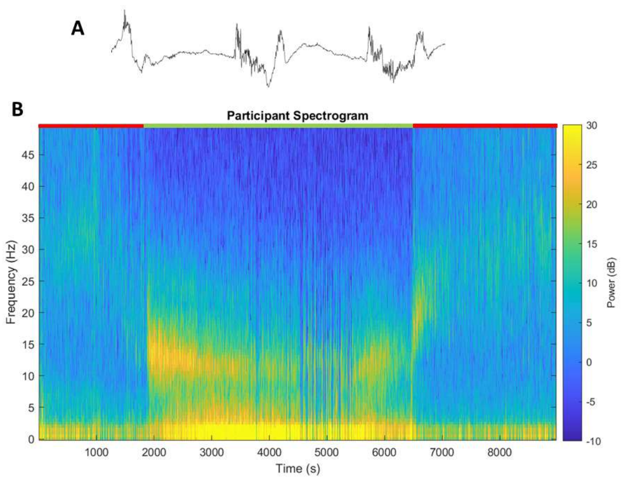

Fig. 1. Visual representations of burst suppression activity in EEG in the time domain (A) and the time-frequency domain demonstrated using a spectrogram with amplitude values in decibels (B). Burst suppression is a unique metabolic state characterized in the EEG signal as alternating patterns of transient bursting activity and isoelectricity. In the spectrogram, they can be identified as vertical lines, indicating uniform amplitude across frequencies. Panel B shows spectrogram for one participant in the dataset. The red bar at the top of the spectrogram indicates moments in the anesthesia procedure where the participant is conscious, and the green bar indicates moments where the participant is unconscious. Burst suppression samples can be located around the 5000s time mark.

patient enters a comatose state, misclassification at this level can result in severe consequences. Thus far, research in the development of a consciousness classifier has been promising [9]–[12]. Relevant to this paper, [13] has developed a useful logistic regression classifier using low dimensional features from EEG time-frequency signals. Specifically, these features include principal component weights via principal components analysis (PCA) of EEG time samples and band-wise power (BWP) of samples along frequency domains. In testing data from operating rooms, logistic regression classifiers utilizing these features produced AUCs of 0.887 and 0.882 respectively. Despite this, isolating classifier performance to burst suppression samples yields below-chance hit rates (Table 1). This is mainly due to the fact that there are less burst suppression samples relative to all other observations. As a result, they are highly prone to misclassification.

TABLE I
STATISTICS FOR CLASSIFIER PERFORMANCE ON TRAINING SETS

	AUC	*Precision*	*Recall*	*F Score*	*Burst Supp. Hit Rate*
BWP + LR	0.944	89.56%	86.91%	88.22%	40.62%
PCA + LR	0.961	91.91%	89.49%	90.68%	43.30%
BWP + SVM	0.985	94.96%	93.60%	94.27%	93.30%
PCA + SVM	0.988	95.35%	94.57%	94.96%	95.98%
BWP + ANN	0.961	91.33%	89.05%	89.90%	50%
PCA + ANN	0.969	93.19%	90.01%	91.57%	63.84%

Our research looked to build upon the classifier developed by [13] by substituting the logistic regression with different supervised machine learning (ML) approaches. Specifically, we utilized support vector machines (SVM) with a radial basis kernel function and single-layer artificial neural networks (ANN). Because these approaches have the potential to generate nonlinear classification boundaries in the feature space, we hypothesized that they would perform as well or better than the logistic regression classifiers with robust performance on burst suppression samples.

II. METHODS

Our dataset consisted of EEG data from 10 participants undergoing a simple anesthesia delivery paradigm [14]. Data was collected in states of consciousness and unconsciousness (Figure 1B) and was grouped into their corresponding conditions. Samples were converted into the time-frequency domain via a multitaper spectral estimation [15] with window length T=2s with no overlap, normalized half-bandwidth TW = 3, and a spectral resolution of 2 Hz [13].

In accordance with the classifier we were replicating, BWP and PCA features were extracted from the time-frequency data. Frequency bands used to generate BWP were slow (0-1 Hz), delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-25 Hz), and gamma (25-50 Hz). Thus BWP constituted a feature vector at time point n of $\boldsymbol{x}_n^{(BWP)} = \mathbf{b}_n = [b_{n,s}, b_{n,\delta}, b_{n,\theta}, b_{n,\alpha}, b_{n,\beta}, b_{n,\gamma}]$. To extract PCA features, the entire data set in the time-frequency domain was subjected to PCA, thus generating eigenvectors that captured a certain percentage of the overall variance in the dataset. Due to the clinical relevance of the first three principal components (PC) (Figure 2A) and the fact that they captured 92.1% of the total variance (Figure 2B), they were the only PCs utilized to transform the data into a PC space (Figure 2C-E) with weights for a time point n of $\boldsymbol{x}_n^{(PCA)} = \mathbf{p}_n = [p_{n,1}, p_{n,2}, p_{n,3}]$ [13].

Features for each time point along with class labels, 0 for conscious and 1 for unconscious, were fed into a logistic regression (LR) model to generate probabilities which describe the likelihood of an observation matching one of the binary classifications. Using the learned weights, we were able to draw linear boundaries along the PC space to visualize the LR decision boundary (Figure 3).

Because LR classifiers generate linear decision boundaries in a given feature space, they were ineffective at generating an accurate decision boundary for identifying unconsciousness during burst suppression. Figure 3 demonstrates that the clustering patterns of conscious and unconscious samples requires a nonlinear ML approach if burst suppression classification is prioritized. Consequently, we sought to utilize support vector machines (SVM) and single-layer artificial neural networks (ANN). Support vector machines are supervised ML models that project samples into a high-dimensional feature space through some non-linear mapping chosen *a priori*, in our case a radial basis kernel function (Figure 4). A linear decision surface is generated in this high dimensional space, yielding a nonlinear decision boundary in the original feature space [16]. Artificial neural networks are a class of supervised ML that utilize weights and biases into an activation function to generate probability values, similar to LR classifiers. Consequently,

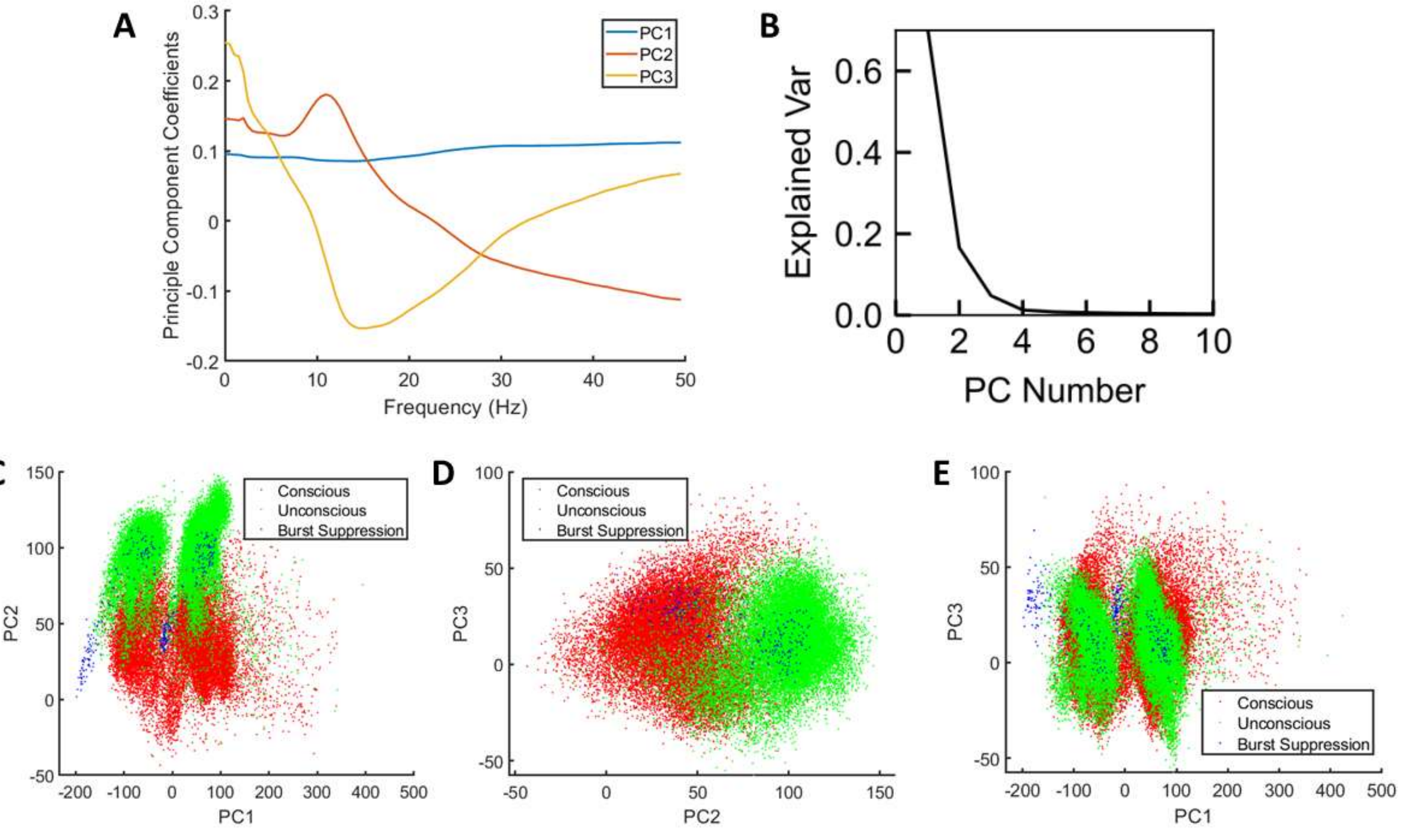

Fig. 2. Summary of principal components analysis of spectrograms from all participants. (A) Principal components 1, 2, and 3 were plotted against spectrogram frequencies to visualize clinical interpretations of the PCs. PC1 reflects DC bias in the data, PC2 reflects differences between slow, delta, theta, and alpha waves from gamma waves, and PC3 reflects differences between slow, delta, and gamma waves from beta waves. (B) The first three PCs accounted for over 92% of the variance in the data, and all PCs passed the third accounted for less than 2%. [13]. (C) Scatter plot of PC1 weights against PC2 weights for all samples in the training data. (D) Scatter plot of PC2 weights against PC3 weights for all samples in the training data. (E) Scatter plot of PC1 weights against PC3 weights for all samples in the training data.

in single-layer ANNs, the network weights can be used to plot linear decision boundaries equal in number to the number of neurons in the layer. We utilized a 10-fold cross validation step to optimize for the number of neurons in the ANNs for each feature. Five neurons performed best for both feature spaces. Consequently, we plotted all five decision boundaries onto the PC space for visualization (Figure 5).

III. Results

Results have been summarized in Table 1. When classifier performance is considered generally along the consciousness and unconsciousness dimension, all six classifiers performed well, with superior performance coming from the SVM classifiers (Figure 6). The lowest AUC calculation was 0.944 when BWP + LR was interrogated for performance on the training set, with the highest AUC being 0.988 by PCA + SVM. When considering classifier accuracy, recall, precision, and F1 scores were utilized. Every classifier performed above chance in all categories, with SVM + PCA outperforming in every category with a recall score of 0.9457, a precision score of 0.9535, and an F1 score of 0.9496.

To isolate performance on burst suppression, we calculated burst suppression hit rate. This was done by dividing the number of burst suppression samples that were predicted to be unconscious by the total number of burst suppression samples in the training set. Classifiers utilizing LR had lower burst suppression hit rate values, while classifiers utilizing SVM had the highest values. The best performing classifier, PCA + SVM, had a burst suppression hit rate of 95.98%, correctly labeling 212 of the 224 burst suppression samples in the training set.

Because the PCA + SVM classifier outperformed all other classifiers in every parameter, particularly burst suppression hit rate (Table 1), we applied the PCA + SVM classifier trained on the training data (70% of all samples) to the test data(30% of all samples). The classifier correctly labeled 84 of the 97 burst suppression samples as unconscious (Burst suppression hit rate = 86.60%) and generated an F1 score of 92.16% while maintaining an AUC of 0.9471. Despite this, from analyzing the decision boundary of the classifier (Figure 4) and the distribution of the samples in PC space (Figure 2C-E), we ultimately concluded that the classifier was overfit to

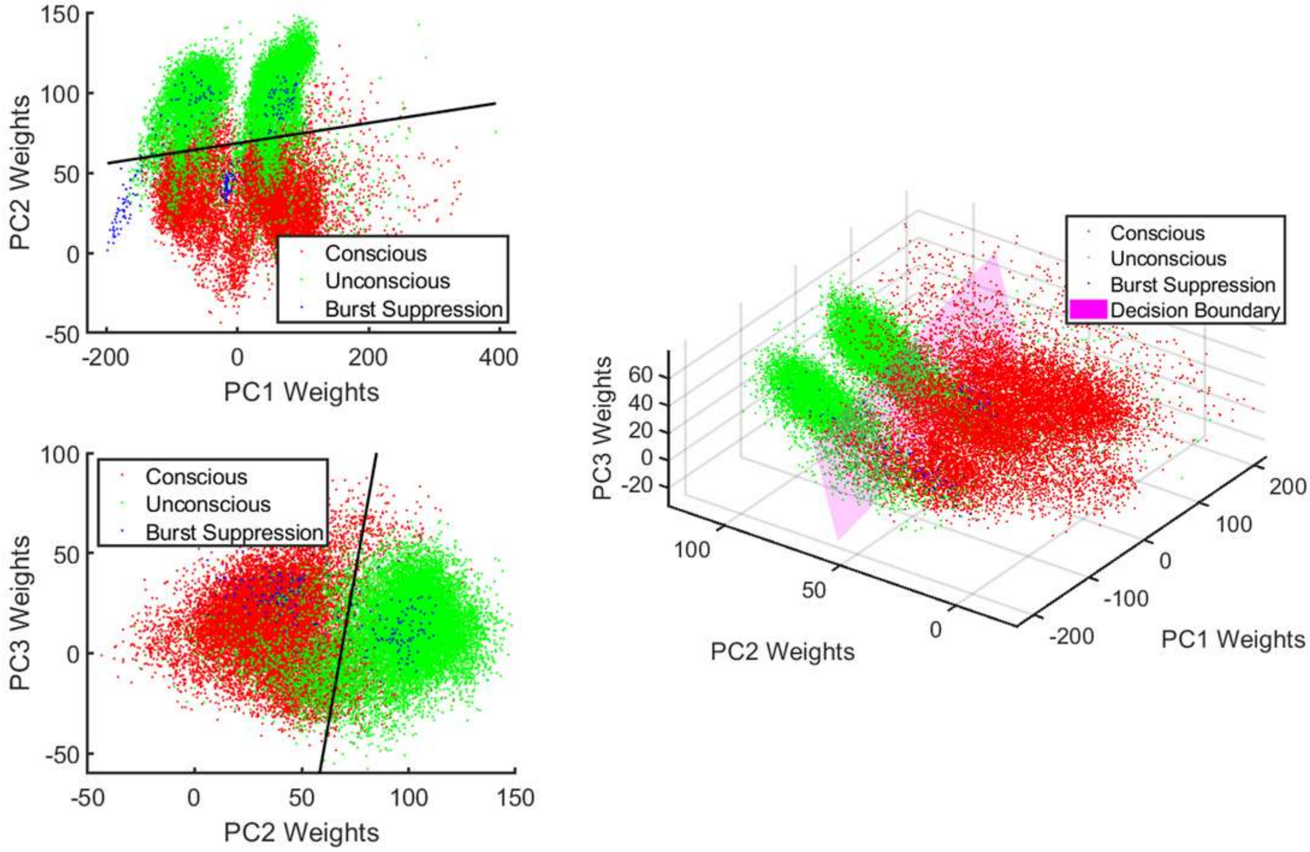

Fig. 3. Decision boundary for PCA + LR. Boundary generated using the beta values from LR. Due to the linear nature of the hyperplane, this classifier is not able to capture space where burst suppression samples occupy. This is mainly due to the fact that linear classification boundaries are not suitable to properly parse the data and largely consider burst suppression points as outliers. As a result, burst suppression data are prone to misclassification using this ML approach.

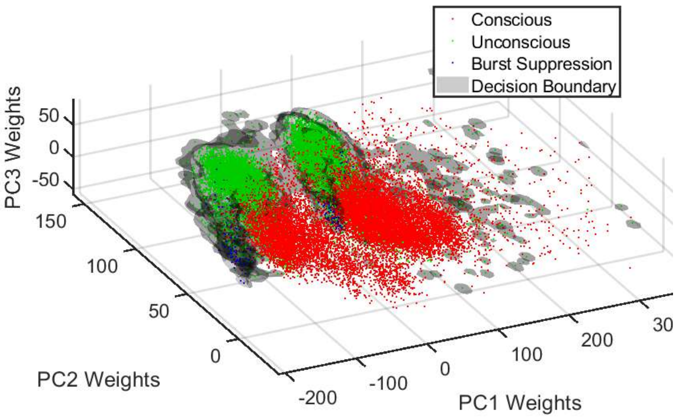

Fig. 4. Decision boundary for PCA + SVM. Boundary generated using support vectors and the calculated with a radial basis kernel function. Unlike the LR classifiers, SVM is capable of independently capturing spaces that are predominantly occupied by burst suppression samples. Despite this, the classifier is overfit to the specific participants in the training data. Because both of the training and testing data composed of the same participants, there was no significant difference in their sampling means along PC1, however it is likely that if novel subjects were projected onto this PC space, PCA + SVM would be highly prone to misclassification generally as well as in burst suppression samples.

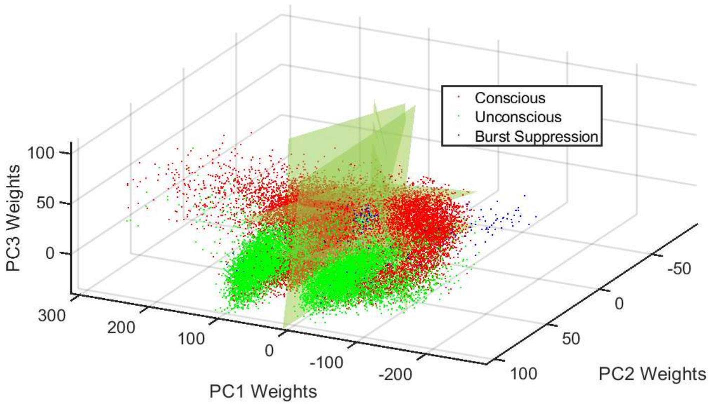

Fig. 5. Hyperplanes associated with hidden layer weights for PCA + ANN. Artificial neural networks learn weights that allow for the generation of linear hyperplanes, similar to LR classifiers, equal in number to the number of neurons in the hidden layer. For this project, five neurons were utilized, thus five boundaries were created. Consequently, 'chambers' of classification can be observed, and the PCA + ANN performed better than chance because is was able to independently capture a cluster of burst suppression samples around PC1 = 0. Despite this, a large cluster of points in the negative side of the PC1 axis was placed in a predominantly conscious classification space, leading to lowered performance overall.

our dataset.

IV. Discussion

Development of a CLAD system requires that EEG data is recorded and readily interpreted by a general classifier. This necessitates a classifier that was trained on a dataset that is reflective of EEG signatures of subjects in unconscious states. The dataset that we utilized has significant differences in the time-frequency values between participants as a result of a DC offset. This can be seen in Figure 3C-E along PC1 weights. There exist two areas of subject data clusters, both averaging PC1 along different values. Because PC1 clinically relates to the baseline DC power associated with a given trial, variability between participants significantly influences the placement of samples in the PC1 domain. Ultimately, because the testing set was comprised of the same participants as the training set, we did not see a significant reduction in performance despite overfitting.

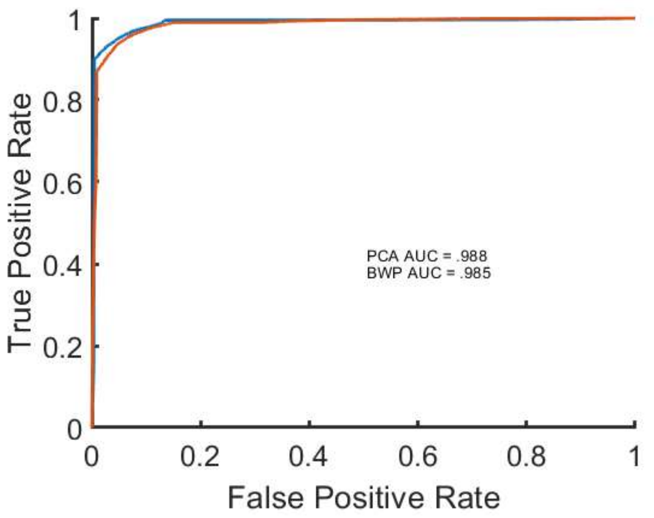

Fig. 6. Receiver operating plot (ROC) for both SVM classifiers with area under the curve (AUC) measurements. Classifier utilizing BWP features is designated by the orange curve and the classifier utilizing PCA features is designated the by the blue curve. The PCA + SVM classifier outperformed all other classifiers.

This fact presents a significant engineering challenge to utilizing the approach we adopted. PC1 is the best dimension to differentiate conscious samples from burst suppression samples (Figure 2C-D), however it is a less effective tool in classification if the data is not normalized properly. While standard normalization approaches offer a promising solution to this issue, such approaches are not well suited for building classifiers that can be utilized in real time. Consequently, this project highlights a key area of further research: the development of a normalization approach that mean centers PC1 for all participants online.

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